

An Interactive Introduction to L^AT_EX

Part 1: The Basics

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August 6, 2026

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1 Why Use L^AT_EX?

L^AT_EX is a document-preparation system designed to produce well-formatted documents. It is especially useful for mathematics and scientific writing. It was created by scientists for scientists, has a large and active community, and can be extended through thousands of packages. These packages support many kinds of work, including papers, presentations, spreadsheets, and other technical documents.

The main idea is to describe *what the content is* rather than manually controlling *how every part looks*. The writer can therefore focus on the content and allow L^AT_EX to manage the formatting.

2 How L^AT_EX Works

A L^AT_EX document is written as plain text containing commands that describe the structure and meaning of the content. The L^AT_EX program processes the source file and produces a formatted document, usually as a PDF.

For example, the following source code emphasizes one word:

```
The rain in Spain falls \emph{mainly} on the plain.
```

Its output is:

The rain in Spain falls *mainly* on the plain.

Commands can also create lists, figures, equations, and many other document elements.

3 A Minimal L^AT_EX Document

A minimal L^AT_EX document has the following form:

```
\documentclass{article}
```

```
\begin{document}
```

```
Hello World! % Your content goes here.
```

```
\end{document}
```

Important ideas are:

- A command normally begins with a backslash, such as `\documentclass`.
- Every document starts with a `\documentclass` command.

- The argument inside curly braces tells L^AT_EX which kind of document to create. In this example, the class is `article`.
- The main content is written between `\begin{document}` and `\end{document}`.
- A percent sign starts a comment. L^AT_EX ignores the remainder of that source-code line.

4 Writing and Compiling the Document

An online editor such as Overleaf can be used to write and compile a L^AT_EX document. It compiles the source automatically and displays the result. The examples in this document can be tested by typing them into a `.tex` file and compiling it.

5 Typesetting Ordinary Text

For most ordinary text, the writer can type normally between `\begin{document}` and `\end{document}`.

Words are separated by one or more spaces. However, several spaces in the source are normally collapsed into a single space in the output.

Paragraphs are separated by one or more blank lines. For example, the following source contains two paragraphs:

```
This is the first paragraph.
```

```
This is the second paragraph.
```

A line break in the source does not necessarily create a line break in the final PDF. For example,

```
The rain in Spain  
falls mainly on the plain.
```

is normally typeset as one continuous sentence.

6 Quotation Marks and Special Characters

6.1 Quotation Marks

Quotation marks require special source-code forms. A left single quotation mark is written with a backtick, and a right single quotation mark is written with an apostrophe:

`'single quotation'`

Double quotation marks are written with two backticks on the left and two apostrophes on the right:

`“double quotation”`

The resulting output is ‘single quotation’ and “double quotation”.

6.2 Special Characters

Some characters have special meanings in $\text{\LaTeX}$. Common examples are:

<code>%</code>	starts a comment,
<code>#</code>	is used with macro parameters,
<code>&</code>	marks an alignment point,
<code>\$</code>	begins or ends inline mathematics.

To display these characters as ordinary text, they are usually escaped with a backslash:

`\% \# \& \$`

The output is: % # & \$.

7 Basic Commands and Environments

7.1 Emphasis

The command `\emph` emphasizes text:

This word is `\emph{important}`.

Output: This word is *important*.

7.2 A Bulleted List

The `itemize` environment creates a bulleted list:

```
\begin{itemize}
  \item Tea
  \item Milk
  \item Biscuits
\end{itemize}
```

The output is:

- Tea
- Milk
- Biscuits

7.3 A Figure

A figure can be inserted with the `figure` environment and the `\includegraphics` command:

```
\begin{figure}
  \centering
  \includegraphics{gerbil}
\end{figure}
```

The following compiling example displays `gerbil.png` when that file is available; otherwise, it displays a placeholder:

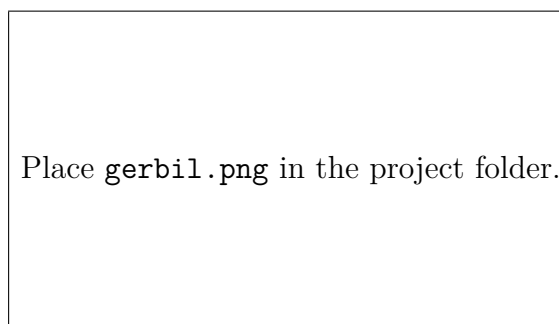


Figure 1: An example of the `figure` environment.

7.4 A Numbered Equation

The `equation` environment displays and numbers an equation:

$$\alpha + \beta + 1 \tag{1}$$

8 Handling Errors

A compilation error occurs when $\text{\LaTeX}$ cannot understand part of the source. For example, misspelling `\emph` as `\meph` causes an “undefined control sequence” error because $\text{\LaTeX}$ does not know the command `\meph`.

Useful advice for handling errors is:

1. Do not panic; errors are a normal part of writing source code.
2. Fix errors as soon as they appear. The most recently added source is often the best place to begin checking.
3. When several errors are reported, start with the first one. A later error may be only a consequence of an earlier mistake.
4. Check the lines above the reported location because the real cause may appear earlier.

9 Typesetting Exercise 1: Text and Special Characters

The following paragraph contains dollar signs, a percentage sign, and quotation marks, so the special characters must be written correctly:

In March 2006, Congress raised that ceiling an additional \$0.79 trillion to \$8.97 trillion, which is approximately 68% of GDP. As of October 4, 2008, the “Emergency Economic Stabilization Act of 2008” raised the current debt ceiling to \$11.3 trillion.

A suitable source form is:

```
In March 2006, Congress raised that ceiling an additional
\$0.79 trillion to \$8.97 trillion, which is approximately
68\% of GDP. As of October 4, 2008, the
‘‘Emergency Economic Stabilization Act of 2008’’
raised the current debt ceiling to \$11.3 trillion.
```

10 Inline Mathematics

Dollar signs are used in pairs to mark mathematics inside a sentence. For example:

Let a and b be distinct positive integers, and let $c = a - b + 1$.

The output is: Let a and b be distinct positive integers, and let $c = a - b + 1$.

Mathematical expressions should be placed in math mode. Compare the ordinary-text expression $c = a - b + 1$ with the properly typeset expression $c = a - b + 1$.

L^AT_EX manages mathematical spacing automatically. Therefore,

`$y=mx+b$`

and

`$y = m x + b$`

produce essentially the same mathematical spacing: $y = mx + b$.

Inline mathematics can also be written with the `math` environment:

```
\begin{math}
y=mx+b
\end{math}
```

11 Superscripts, Subscripts, and Grouping

A caret creates a superscript, and an underscore creates a subscript:

$$y = c_2x^2 + c_1x + c_0.$$

Curly braces group several characters into one superscript or subscript. For example, the correct Fibonacci relation is

$$F_n = F_{n-1} + F_{n-2}.$$

Without braces, `F_n-1` means $F_n - 1$, not F_{n-1} .

12 Greek Letters and Common Mathematical Notation

Greek letters are written with commands such as `\alpha`, `\beta`, `\mu`, `\Omega`, and `\omega`.

For example,

$$\mu = Ae^{Q/RT},$$

where μ , A , Q , R , and T are mathematical symbols.

A summation is written with `\sum`:

$$\Omega = \sum_{k=1}^n \omega_k.$$

Here, $k = 1$ is the lower limit and n is the upper limit.

13 Displayed Equations

A large mathematical expression can be displayed on its own line with the `equation` environment. The roots of a quadratic equation are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad (2)$$

where a , b , and c are the coefficients of the quadratic equation.

The commands used above include:

- `\frac{numerator}{denominator}` for a fraction,
- `\pm` for the plus-or-minus symbol,
- `\sqrt{...}` for a square root,
- `^` for superscripts.

Blank lines should not be placed inside a mathematical environment.

14 Understanding Environments

An environment is a context created with matching `\begin` and `\end` commands:

```
\begin{environment-name}  
    Content  
\end{environment-name}
```

The same mathematical source can appear differently in different contexts. For example, the expression $\Omega = \sum_{k=1}^n \omega_k$ appears compactly in a sentence, while

$$\Omega = \sum_{k=1}^n \omega_k \quad (3)$$

appears larger in a displayed equation. The summation symbol and the positions of its limits change according to the context.

15 List Environments

The `itemize` environment creates bullet points:

- Biscuits
- Tea

The `enumerate` environment creates numbered items:

1. Biscuits
2. Tea

Their source forms are:

```
\begin{itemize}
  \item Biscuits
  \item Tea
\end{itemize}
```

```
\begin{enumerate}
  \item Biscuits
  \item Tea
\end{enumerate}
```

16 Packages and the Preamble

Many basic commands and environments are built into $\text{\LaTeX}$. Packages are libraries that provide additional commands and environments. Packages are loaded in the preamble, which is the part of the source between `\documentclass` and `\begin{document}`.

For example, the `amsmath` package from the American Mathematical Society is loaded as follows:

```
\documentclass{article}
\usepackage{amsmath}

\begin{document}

% Commands from amsmath can now be used here.

\end{document}
```

This document also loads `graphicx` so that images can be included.

17 Unnumbered Equations with `amsmath`

The starred `equation*` environment displays an equation without assigning an equation number:

$$\Omega = \sum_{k=1}^n \omega_k.$$

Its source is:

```
\begin{equation*}
  \Omega=\sum_{k=1}^n\omega_k
\end{equation*}
```

18 Mathematical Operators

In mathematics, adjacent letters are normally interpreted as separate variables multiplied together. Therefore, named operators should use appropriate operator commands.

A minimization expression should use `\min`:

$$\min_{x,y} \left\{ (1-x)^2 + 100(y-x^2)^2 \right\}.$$

For operators without a predefined command, `\operatorname` can be used. For example,

$$\beta_i = \frac{\operatorname{Cov}(R_i, R_m)}{\operatorname{Var}(R_m)}.$$

In this expression, `Cov` denotes covariance and `Var` denotes variance.

19 Aligned Equations

The `align*` environment aligns a sequence of equations. The ampersand marks the alignment position, and a double backslash starts a new line:

$$\begin{aligned} (x+1)^3 &= (x+1)(x+1)(x+1) \\ &= (x+1)(x^2+2x+1) \\ &= x^3+3x^2+3x+1. \end{aligned}$$

The corresponding source is:

```

\begin{align*}
(x+1)^3 &= (x+1)(x+1)(x+1) \\
&= (x+1)(x^2+2x+1) \\
&= x^3+3x^2+3x+1
\end{align*}

```

Because the environment name contains a star, the equations are not numbered.

20 Typesetting Exercise 2: Probability and Mathematics

Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed random variables with $E[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2 < \infty$, and let

$$S_n = \frac{1}{n} \sum_{i=1}^n X_i$$

denote their mean. Then, as n approaches infinity, the random variables

$$\sqrt{n}(S_n - \mu)$$

converge in distribution to a normal distribution $\mathcal{N}(0, \sigma^2)$.

The command for the infinity symbol is `\infty`. The source for the main statement is:

```

Let $X_1, X_2, \ldots, X_n$ be independent and identically
distributed random variables with
 $E[X_i] = \mu$  and
 $\text{Var}[X_i] = \sigma^2 < \infty$ , and let

```

```

\begin{equation*}
S_n = \frac{1}{n} \sum_{i=1}^n X_i
\end{equation*}

```

```

denote their mean. Then, as $n$ approaches infinity,
 $\sqrt{n}(S_n - \mu)$  converges in distribution to
 $\mathcal{N}(0, \sigma^2)$ .

```

21 Summary

This document has demonstrated how to:

- create a minimal L^AT_EX article;

- write and format ordinary text;
- use comments, commands, arguments, and special characters;
- write correct quotation marks;
- emphasize text;
- create figures, bulleted lists, and numbered lists;
- recognize and respond to compilation errors;
- write inline and displayed mathematics;
- use superscripts, subscripts, grouping, Greek letters, fractions, square roots, sums, and infinity;
- work with environments;
- load packages in the preamble;
- create numbered and unnumbered equations;
- use mathematical operators;
- align multi-line equations.

A later stage of learning L^AT_EX can introduce more structured documents with sections, cross-references, figures, tables, and bibliographies.